Composite Quantum Systems and Tensor Products

Hilbert Space Construction and Operator Extension

Abstract

Composite quantum systems are formed by combining multiple subsystems, with their total Hilbert space constructed via the tensor product of individual Hilbert spaces. This document provides a comprehensive treatment of tensor product construction, basis formation, operator extension, and key mathematical properties, with applications to entanglement and quantum information processing.

Contents

1 Introduction

A *composite quantum system* consists of two or more subsystems that interact or exist together. Unlike classical systems, quantum composite systems exhibit *entanglement*, where the joint state cannot be factored into individual subsystem states.

2 Hilbert Space Construction

2.1 Definition of Composite Hilbert Space

Definition 1 (Composite Hilbert Space)**.** *Given subsystems with Hilbert spaces $\mathcal{H}_A$ (dimension d_A) and $\mathcal{H}_B$ (dimension d_B), the composite Hilbert space is:*

$$\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B \tag{1}$$

with dimension $\dim(\mathcal{H}_{AB}) = d_A \cdot d_B$*.*

2.2 Tensor Product Axioms

The tensor product satisfies:

1. **Bilinearity**: $(\alpha|\psi_1\rangle + \beta|\psi_2\rangle) \otimes |\phi\rangle = \alpha|\psi_1\rangle \otimes |\phi\rangle + \beta|\psi_2\rangle \otimes |\phi\rangle$
2. **Symmetry**: $|\psi\rangle \otimes |\phi\rangle = |\phi\rangle \otimes |\psi\rangle$ (up to isomorphism)
3. **Universality**: $\mathcal{H}_A \otimes \mathcal{H}_B$ is universal for bilinear maps

3 Basis Construction

3.1 Product Basis Theorem

Theorem 1 (Product Basis)**.** *If $\{|i\rangle_A\}_{i=1}^{d_A}$ and $\{|j\rangle_B\}_{j=1}^{d_B}$ are orthonormal bases for $\mathcal{H}_A$ and $\mathcal{H}_B$, then $\{|i\rangle_A \otimes |j\rangle_B\}_{i,j}$ forms an orthonormal basis for $\mathcal{H}_{AB}$.*

Proof. **Linearity**: Product vectors span $\mathcal{H}_{AB}$ by bilinearity.
Orthogonality:

$$\begin{aligned} \langle i_A j_B | k_A l_B \rangle &= \langle i_A | k_A \rangle \langle j_B | l_B \rangle \\ &= \delta_{ik}\delta_{jl} \end{aligned} \tag{2}$$

Completeness: $\dim(\mathcal{H}_{AB}) = d_A d_B$ matches basis size. □

3.2 Notation Conventions

Standard:	$\|ij\rangle_{AB} \equiv \|i\rangle_A \otimes \|j\rangle_B$
Physics:	$\|i\rangle_A\|j\rangle_B$, $\|i,j\rangle$
QIP:	$\|ij\rangle$ (implicit subsystems)

3.3 Two-Qubit Example

For qubits ($\mathcal{H}_A = \mathcal{H}_B = \mathbb{C}^2$):

$$\begin{gathered} \mathcal{H}_{AB} = \mathbb{C}^2 \otimes \mathbb{C}^2 = \mathbb{C}^4 \\ \text{Basis: } \{|00\rangle, |01\rangle, |10\rangle, |11\rangle\} \end{gathered} \tag{3}$$

4 Tensor Product Properties

4.1 Main Mathematical Properties

1. **Dimension**: $\dim(A \otimes B) = \dim(A)\dim(B)$
2. **Orthonormality preservation**:

$$\langle \psi_1 | \phi_1 \rangle \langle \psi_2 | \phi_2 \rangle = \langle \psi_1 \otimes \psi_2 | \phi_1 \otimes \phi_2 \rangle \tag{4}$$

3. **Unitary embedding**: $U_A \otimes I_B$, $I_A \otimes U_B$
4. **Associativity**: $(A \otimes B) \otimes C \cong A \otimes (B \otimes C)$

4.2 Partial Inner Product

For $|\Psi\rangle_{AB} = \sum_{ij} c_{ij} |i\rangle_A |j\rangle_B$:

$$\langle \phi_A | \Psi \rangle_{AB} = \sum_i \langle \phi_A | i \rangle_A \sum_j c_{ij} |j\rangle_B \in \mathcal{H}_B \tag{5}$$

5 Operator Extension

5.1 Tensor Product of Operators

Definition 2 (Operator Tensor Product)**.** *For $\hat{A} \in \mathcal{B}(\mathcal{H}_A)$, $\hat{B} \in \mathcal{B}(\mathcal{H}_B)$:*

$$\hat{A} \otimes \hat{B} : \mathcal{H}_A \otimes \mathcal{H}_B \to \mathcal{H}_A \otimes \mathcal{H}_B \tag{6}$$

defined by $(\hat{A} \otimes \hat{B})(|\psi\rangle \otimes |\phi\rangle) = \hat{A}|\psi\rangle \otimes \hat{B}|\phi\rangle$.

5.2 Properties of Operator Tensor Product

Lemma 1 (Operator Tensor Properties)**.** *(a)* ***Linearity:*** $(\alpha \hat{A}_1 + \beta \hat{A}_2) \otimes \hat{B} = \alpha \hat{A}_1 \otimes \hat{B} + \beta \hat{A}_2 \otimes \hat{B}$

(b) ***Mixed terms:*** $\hat{A}_1 \otimes \hat{B}_1 + \hat{A}_2 \otimes \hat{B}_2$

(c) ***Adjoint:*** $(\hat{A} \otimes \hat{B})^\dagger = \hat{A}^\dagger \otimes \hat{B}^\dagger$

(d) ***Unitary:*** *$\hat{U}_A \otimes \hat{U}_B$ is unitary if $\hat{U}_A$, $\hat{U}_B$ unitary*

5.3 Local vs Global Operators

Type	**Form**	**Action**
Local A	$\hat{A} \otimes I_B$	Acts only on subsystem A
Local B	$I_A \otimes \hat{B}$	Acts only on subsystem B
Separable	$\hat{A} \otimes \hat{B}$	Product operator
Entangling	General $\hat{C}_{AB}$	Correlates subsystems

Table 1: Operator Classification

6 Explicit Matrix Construction

6.1 Two-Qubit Pauli Operators

For basis $\{|0\rangle, |1\rangle\}$:

$$\begin{aligned} \sigma_x^A = \sigma_x \otimes I &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \otimes \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \end{aligned} \tag{7}$$

6.2 General Construction Algorithm

1. Choose bases $\{|i\rangle_A\}$, $\{|j\rangle_B\}$
2. Matrix elements: $\langle k\ell|\hat{A} \otimes \hat{B}|ij\rangle = A_{ki}B_{\ell j}$
3. Kronecker product ordering: $A \otimes B = \begin{bmatrix} a_{11}B & a_{12}B & \cdots \\ a_{21}B & a_{22}B & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}$

7 Entanglement and Non-Separability

7.1 Separable vs Entangled States

Definition 3. $|\Psi\rangle_{AB}$ *is* separable *if:*

$$|\Psi\rangle_{AB} = |\psi\rangle_A \otimes |\phi\rangle_B \tag{8}$$

Otherwise, $|\Psi\rangle_{AB}$ *is* entangled.

7.2 Bell State Example

Maximal entanglement in two qubits:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \notin \text{span}\{|\psi\rangle \otimes |\phi\rangle\} \tag{9}$$

7.3 Reduced Density Matrix

Tracing over subsystem B:

$$\rho_A = \text{Tr}_B(|\Psi\rangle\langle\Psi|) = \sum_j \langle j|_B|\Psi\rangle\langle\Psi||j\rangle_B \tag{10}$$

8 Multi-Partite Systems

8.1 Iterative Tensor Products

For N subsystems:

$$\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2 \otimes \cdots \otimes \mathcal{H}_N \tag{11}$$

Basis: $\{|i_1 i_2 \cdots i_N\rangle\}$ where $|i_1 i_2 \cdots i_N\rangle = |i_1\rangle_1 \otimes \cdots \otimes |i_N\rangle_N$.

8.2 Partial Trace Generalization

For subsystems $S \subset \{1, \ldots, N\}$:

$$\rho_S = \mathrm{Tr}_{\overline{S}} \left(\sum_{i,j} |i\rangle\langle j|\rho|i\rangle\langle j| \right) \tag{12}$$

9 Physical Applications

9.1 Quantum Information Processing

- **Quantum circuits**: Gates as $U_A \otimes I_B$ or entangling gates
- **Teleportation**: Bell states in $\mathcal{H}_{AB}$
- **Superdense coding**: 4^d states in d-qubit space

9.2 Quantum Error Correction

Logical qubits encoded in $\mathcal{H}^{\otimes n}$, error operators act locally.

10 Key Insights

1. **Dimensional explosion**: n qubits $\to 2^n$ dimensions
2. **Entanglement geometry**: Non-factorizable states span full space
3. **Local control**: $U_A \otimes I_B$ cannot create entanglement
4. **Measurement correlations**: $\langle AB\rangle \neq \langle A\rangle\langle B\rangle$

11 Conclusion

The tensor product construction provides the mathematical foundation for composite quantum systems. It naturally accommodates entanglement—the defining feature of quantum mechanics—while preserving locality through operator tensor products. This structure underpins quantum computing, quantum communication, and quantum many-body physics.